Supplementary Material

Endpoint-Calibrated Spatial Graph Rollouts for Breast DCE-MRI FTV Forecasting

Iris Seaman, Tamer Oraby, and Murad Moqbel

This supplement collects additional ablation, calibration, scalar-baseline, source-robustness, and external stress-test evidence supporting the manuscript. All source-cohort rows use the same patient-level fold isolation, residual-pool exclusion, and conformal calibration protocol described in the main paper. Patient-level prediction rows and raw imaging data are not included here.

S1. Endpoint-Loss Calibration Ablations

Table 1: T0-to-T3 endpoint-loss calibration ablations. Negative deterministic bias indicates underprediction.

Model	Det MAE	Det bias	MC MAE	Raw 90% cov.	Conf. width	CRPS	Alive err.	High-T3 MAE
Baseline	12.88	-12.51	15.26	0.766	59.51	8.35	40.78	42.59
FTV .020 + alive .005	5.88	-1.73	7.61	0.876	27.95	5.16	2.69	29.65
FTV only	5.99	-1.51	7.31	0.875	26.92	5.15	7.00	30.85
FTV .010 + alive .002	5.99	-1.50	7.53	0.883	29.14	5.19	2.70	29.79

S2. Graph-Neighborhood and Edge-Attribute Ablations

The graph-family search includes no-edge, spatial, radial, feature-only, hybrid spatial-feature, and radial-biologic edge variants. These rows are kept in the supplement because they support the retained model choice without making the main manuscript table too wide.

Table 2: Full T0-to-T3 graph-family ablation search on the source cohort.

Model	n	MC MAE	CRPS	Raw 90% cov.	Raw width	Conf. width	SWD	Chamfer	Dice
Radial-bio k=8	758	5.52	3.95	0.872	20.05	23.29	1.18	3.46	0.052
Hybrid-bio k=8	758	5.56	4.00	0.877	19.21	22.38	1.17	3.45	0.053
Radial-bio k=16	758	5.61	4.13	0.877	20.63	24.01	1.18	3.46	0.052
Spatial-bio k=4	758	5.67	4.01	0.868	19.50	23.05	1.19	3.47	0.052
Radial-geo k=8	758	5.71	4.13	0.865	19.52	23.31	1.18	3.46	0.052
Hybrid-bio alpha=.75 k=8	758	5.84	4.10	0.868	19.66	23.00	1.18	3.45	0.053
Radial-bio k=4	758	5.99	4.24	0.879	21.10	24.25	1.17	3.46	0.052
Radial k=8	758	6.25	4.38	0.864	21.06	26.01	1.17	3.45	0.052
No edges	758	6.59	4.56	0.871	22.87	26.50	1.21	3.49	0.051
Feature-volume k=8	758	7.23	4.91	0.873	24.06	27.34	1.18	3.46	0.052
FTV .010 only	758	7.31	5.15	0.875	23.96	26.92	1.21	3.50	0.051
Hybrid alpha=.50 k=8	758	7.35	5.00	0.864	24.00	28.98	1.20	3.48	0.052
Feature-all k=8	758	7.35	4.99	0.873	24.02	28.20	1.18	3.46	0.052
Hybrid alpha=.25 k=8	758	7.36	4.98	0.864	24.21	28.55	1.19	3.47	0.052
Hybrid alpha=.75 k=8	758	7.43	5.04	0.864	24.26	30.13	1.20	3.48	0.052
Feature PE/SER k=8	758	7.47	5.05	0.860	24.76	29.28	1.17	3.45	0.052
Spatial k=16	758	7.48	5.19	0.880	25.12	28.57	1.20	3.48	0.051
FTV .010 + alive .002	758	7.53	5.19	0.883	24.81	29.14	1.21	3.49	0.051
Spatial k=4	758	7.53	5.04	0.864	24.54	30.22	1.21	3.49	0.051
Endpoint-calibrated	758	7.61	5.16	0.876	24.42	27.95	1.21	3.49	0.051
Endpoint full k=8	758	7.63	5.17	0.872	24.66	29.20	1.20	3.48	0.051
Original graph	758	15.26	8.35	0.766	56.87	59.51	1.32	3.66	0.047

S3. Scalar, Hybrid, and Temporal Baselines

These scalar-only baselines define the boundary of the graph claim. They use observed FTV history through the conditioning visit and the same residual Monte Carlo and conformal evaluation family. They do not replace the structured tumor-state forecast produced by the graph model.

Table 3: T0-to-T3 scalar, graph, and hybrid residual-MC comparison.

Family	Model	n	Det MAE	MC MAE	Raw 90% cov.	Raw width	CRPS
Hybrid residual MC	Hybrid graph+scalar MC	758	4.48	4.50	0.897	18.20	3.71
Temporal scalar residual MC	Ridge temporal scalar MC	758	4.66	4.70	0.891	18.98	3.84
Scalar residual MC	Last-observed scalar MC	758	4.81	4.82	0.898	18.60	3.76
Temporal scalar residual MC	Log-ridge temporal scalar MC	758	4.82	4.89	0.896	20.11	4.05
Graph residual MC	Graph retained	758	4.63	5.56	0.877	19.21	4.00
Graph residual MC	Graph baseline	758	12.88	15.26	0.766	56.87	8.35
Temporal scalar residual MC	Kernel temporal scalar MC	758	11.93	15.82	0.896	54.21	10.14

Table 4: Additional scalar temporal centers scored with residual Monte Carlo.

Bucket	Model	n	Center MAE	MC MAE	Raw 90% cov.	Raw width	CRPS
T0–T3	RBF temporal log	758	11.93	15.82	0.896	54.21	10.14
T0–T3	Ridge temporal log	758	4.82	4.89	0.896	20.11	4.05
T0–T3	Ridge temporal raw	758	4.66	4.70	0.891	18.98	3.84
T1–T3	RBF temporal log	758	12.27	16.25	0.897	56.65	10.45
T1–T3	Ridge temporal log	758	4.87	4.93	0.893	19.77	4.09
T1–T3	Ridge temporal raw	758	4.78	4.81	0.898	19.15	3.92
T2–T3	RBF temporal log	758	12.10	16.29	0.900	54.88	10.35
T2–T3	Ridge temporal log	758	4.22	4.27	0.897	17.45	3.54
T2–T3	Ridge temporal raw	758	4.51	4.54	0.897	17.18	3.70

S4. Burden-Conditional Calibration

Table 5: Observed-T3-FTV quartile calibration for key graph and scalar readouts.

Stratum	Model	n	Mean obs. FTV	MC MAE	Raw 90% cov.	Conf. 90% cov.	Conf. width	CRPS
Q1 lowest	Graph retained	190	3.33	3.76	0.963	0.968	20.23	1.98
Q1 lowest	Hybrid graph+scalar MC	190	3.33	2.17	0.974	0.974	19.18	1.93
Q1 lowest	Last-observed scalar MC	190	3.33	2.88	0.974	0.974	19.20	2.00
Q1 lowest	Graph baseline	190	3.33	14.44	0.389	0.805	51.08	5.70
Q2	Graph retained	189	8.80	3.01	0.968	0.974	22.83	1.73
Q2	Hybrid graph+scalar MC	189	8.80	2.04	0.989	0.989	19.12	1.60
Q2	Last-observed scalar MC	189	8.80	2.42	0.989	0.989	19.11	1.62
Q2	Graph baseline	189	8.80	13.38	0.847	0.937	53.12	4.77
Q3	Graph retained	189	18.32	3.99	0.915	0.947	23.01	2.70
Q3	Hybrid graph+scalar MC	189	18.32	3.35	0.963	0.963	19.26	2.50
Q3	Last-observed scalar MC	189	18.32	3.45	0.963	0.963	19.15	2.53
Q3	Graph baseline	189	18.32	12.05	0.942	0.974	58.49	4.31
Q4 highest	Graph retained	190	69.19	11.48	0.663	0.716	23.45	9.56
Q4 highest	Hybrid graph+scalar MC	190	69.19	10.41	0.663	0.668	18.88	8.78
Q4 highest	Last-observed scalar MC	190	69.19	10.50	0.668	0.674	19.05	8.86
Q4 highest	Graph baseline	190	69.19	21.15	0.889	0.889	75.33	18.59

Table 6: Tail-stratified T0-to-T3 calibration for key graph and scalar readouts.

Stratum	Model	n	Mean obs. FTV	MC MAE	Raw 90% cov.	Conf. 90% cov.	Conf. width	CRPS
Below top decile (<58.73 mL)	Graph retained	682	14.90	3.97	0.924	0.944	22.30	2.49
Below top decile (<58.73 mL)	Hybrid graph+scalar MC	682	14.90	2.95	0.952	0.953	19.14	2.32
Below top decile (<58.73 mL)	Last-observed scalar MC	682	14.90	3.30	0.955	0.955	19.16	2.36
Below top decile (<58.73 mL)	Graph baseline	682	14.90	12.22	0.771	0.921	56.47	4.94
Top 5% (>=84.88 mL)	Graph retained	38	159.88	31.57	0.237	0.289	22.10	28.62
Top 5% (>=84.88 mL)	Hybrid graph+scalar MC	38	159.88	28.06	0.132	0.132	18.64	25.28
Top 5% (>=84.88 mL)	Last-observed scalar MC	38	159.88	26.82	0.211	0.211	18.85	24.19
Top 5% (>=84.88 mL)	Graph baseline	38	159.88	73.78	0.447	0.447	92.13	64.97
Top decile (>=58.73 mL)	Graph retained	76	115.06	19.89	0.461	0.513	23.03	17.55
Top decile (>=58.73 mL)	Hybrid graph+scalar MC	76	115.06	18.39	0.408	0.408	18.82	16.16
Top decile (>=58.73 mL)	Last-observed scalar MC	76	115.06	18.47	0.395	0.408	18.87	16.31
Top decile (>=58.73 mL)	Graph baseline	76	115.06	42.59	0.724	0.724	86.84	39.04

S5. Subtype Calibration

Table 7: T0-to-T3 retained-model calibration by molecular subtype. Residual calibration was not refit by subgroup.

Subtype	n	Det MAE	MC MAE	Raw 90% cov.	Conf. 90% cov.	Conf. width	CRPS
HR+/HER2+	128	3.29	4.90	0.906	0.922	22.76	3.13
HR+/HER2-	312	4.49	5.24	0.869	0.894	22.14	3.77
HR-/HER2+	57	6.70	7.52	0.912	0.930	22.25	6.01
TNBC	261	5.00	5.86	0.866	0.893	22.50	4.26

S6. Reliability and PIT Aggregates

Table 8: Coverage-vs-nominal reliability for T0-to-T3 graph baseline and retained graph model.

Model	Nominal	Emp. cov.	Mean width	n
Graph baseline	0.500	0.340	14.61	758
Graph baseline	0.600	0.434	19.19	758
Graph baseline	0.700	0.520	26.23	758
Graph baseline	0.800	0.632	36.91	758
Graph baseline	0.900	0.766	56.87	758
Graph baseline	0.950	0.875	95.32	758
Graph retained	0.500	0.482	3.98	758
Graph retained	0.600	0.571	5.29	758
Graph retained	0.700	0.664	7.05	758
Graph retained	0.800	0.770	10.00	758
Graph retained	0.900	0.877	19.21	758
Graph retained	0.950	0.925	34.32	758

Table 9: T0-to-T3 PIT decile counts. Uniformity is approximate because the empirical MC sample is finite and residual-calibrated.

PIT bin	Baseline n	Baseline frac.	Retained n	Retained frac.
0.0-0.1	246	0.325	108	0.142
0.1-0.2	121	0.160	104	0.137
0.2-0.3	82	0.108	84	0.111
0.3-0.4	64	0.084	81	0.107
0.4-0.5	65	0.086	87	0.115
0.5-0.6	52	0.069	64	0.084
0.6-0.7	25	0.033	63	0.083
0.7-0.8	41	0.054	58	0.077
0.8-0.9	35	0.046	49	0.065
0.9-1.0	27	0.036	60	0.079

S7. Source and External Robustness

Table 10: Source-stratified internal robustness for the retained graph model.

Source	n	MC MAE	CRPS	Raw 90% cov.	Conf. width	SWD	Chamfer
ACRIN-6698	202	4.55	2.93	0.881	21.78	1.13	3.31
ISPY2	556	5.93	4.39	0.876	22.59	1.19	3.50

Table 11: Independent Breast-MRI-NACT-Pilot T0-to-T3 stress test on 11 graph-ready external patients.

Model	Det MAE	Det bias	MC MAE	Raw 90% cov.	Conf. 90% cov.
Endpoint+Active	20.17	19.71	23.51	0.455	0.455
No-edge endpoint	19.21	18.82	22.11	0.455	0.455
Radial-biologic k=8	16.70	16.36	19.31	0.455	0.455
Hybrid-Edge k=8	12.40	12.13	15.01	0.545	0.545

S8. Imaging-Burden Readout Checks

Table 12: Exploratory imaging-burden threshold readouts. These are not pathology-response model claims.

Endpoint	Score	n	Event rate	AUC	Sensitivity	Specificity	PPV	NPV
T3 FTV < 5 mL	MC probability P(T3 FTV < 5 mL)	758	0.210	0.957	0.755	0.970	0.870	0.937
T3 FTV < 5 mL	Last observed FTV < 5 mL	758	0.210	0.961	0.679	0.980	0.900	0.920
T3 FTV > 20 mL	MC probability P(T3 FTV > 20 mL)	758	0.325	0.984	0.967	0.914	0.844	0.983
T3 FTV > 20 mL	Last observed FTV > 20 mL	758	0.325	0.986	0.972	0.902	0.827	0.985

S9. Interpretation Boundary

The additional ablations support the retained graph-family design, but they do not change the central interpretation of the paper. Scalar and hybrid readouts remain strong when the only endpoint is scalar FTV. The graph contribution is therefore calibrated structured-state forecasting: the model forecasts a future tumor graph state, active-node state, and FTV distribution, then scalar or hybrid readouts can be layered on top when a scalar-only clinical query is desired. The independent Breast-MRI-NACT-Pilot analysis is a preliminary external stress test, not powered definitive clinical validation.